FURTHER MECHANICS / MISCELLANEOUS MECHANICS

Dimensional Analysis

Worked solutions 10 questions · 48 marks

Instructions

Unless a question states otherwise, take $g = 9.8\,\mathrm{m\,s^{-2}}$ wherever a numerical value of g is required. Solutions also give final answers for $g = 10\,\mathrm{m\,s^{-2}}$ where these differ.

Questions

Question 1	*5 marks*	2
Question 2	*5 marks*	4
Question 3	*6 marks*	6
Question 4	*4 marks*	8
Question 5	*6 marks*	9
Question 6	*6 marks*	11
Question 7	*2 marks*	13
Question 8	*4 marks*	14
Question 9	*6 marks*	15
Question 10	*4 marks*	17

1. **(a)** State the dimensions of force. **[1]**

(b) The braking distance of a vehicle can be modelled by the formula

$$d = m^{\alpha} u^{\beta} R^{\gamma}$$

where

m is the mass of the vehicle in kg
u is the initial speed in $\mathrm{m\,s^{-1}}$
R is the constant braking force in N

Use dimensional analysis to find the values of α, β and γ. **[4]**

Solution

(a) Force is mass $\times$ acceleration, so

$$[F] = \mathrm{M} \times \mathrm{LT}^{-2} = \mathrm{MLT}^{-2}$$

(b) Write the dimensions of each quantity:

$$[d] = \mathrm{L}, \qquad [m] = \mathrm{M}, \qquad [u] = \mathrm{LT}^{-1}, \qquad [R] = \mathrm{MLT}^{-2}$$

Given

$$d = m^{\alpha} u^{\beta} R^{\gamma}$$

the dimensions on the right-hand side are

$$\begin{aligned}[m^{\alpha} u^{\beta} R^{\gamma}] &= \mathrm{M}^{\alpha}(\mathrm{LT}^{-1})^{\beta}(\mathrm{MLT}^{-2})^{\gamma} \\ &= \mathrm{M}^{\alpha}\mathrm{L}^{\beta}\mathrm{T}^{-\beta} \times \mathrm{M}^{\gamma}\mathrm{L}^{\gamma}\mathrm{T}^{-2\gamma} \\ &= \mathrm{M}^{\alpha+\gamma}\mathrm{L}^{\beta+\gamma}\mathrm{T}^{-\beta-2\gamma}\end{aligned}$$

This must equal the dimensions of d, so

$$\mathrm{M}^{\alpha+\gamma}\mathrm{L}^{\beta+\gamma}\mathrm{T}^{-\beta-2\gamma} = \mathrm{L} = \mathrm{M}^{0}\mathrm{L}^{1}\mathrm{T}^{0}$$

Equating powers of M, L and T gives

$$\alpha + \gamma = 0, \qquad \beta + \gamma = 1, \qquad -\beta - 2\gamma = 0$$

The equation $-\beta - 2\gamma = 0$ gives

$$\beta = -2\gamma$$

Substitute into $\beta + \gamma = 1$:

$$\begin{aligned}-2\gamma + \gamma &= 1 \\ -\gamma &= 1 \\ \gamma &= -1\end{aligned}$$

Then

$$\beta = -2(-1) = 2$$

and $\alpha + \gamma = 0$ gives

$$\alpha + (-1) = 0$$

so

$$\alpha = 1$$

Hence

$$\alpha = 1, \qquad \beta = 2, \qquad \gamma = -1$$

2. A small satellite moves in a circular orbit of radius r about the centre of a much more massive planet of mass M. It is suggested that the period T of the orbit depends only on G, M and r, and that

$$T = kG^{\alpha}M^{\beta}r^{\gamma}$$

where k is a dimensionless constant.

It is given that

$$G = 6.67 \times 10^{-11}\,\mathrm{m^3\,kg^{-1}\,s^{-2}}$$

(a) Write down the dimensions of G. **[1]**

(b) By using dimensional analysis, determine the values of α, β and γ. **[4]**

Solution

(a) From

$$G = 6.67 \times 10^{-11}\,\mathrm{m^3\,kg^{-1}\,s^{-2}}$$

the dimensions of G are

$$[G] = \mathrm{M^{-1}L^3T^{-2}}$$

(b) The constant k is dimensionless, so

$$[T] = [G]^{\alpha}[M]^{\beta}[r]^{\gamma}$$

Now

$$[T] = \mathrm{T}, \qquad [G] = \mathrm{M^{-1}L^3T^{-2}}, \qquad [M] = \mathrm{M}, \qquad [r] = \mathrm{L}$$

so

$$\begin{aligned}\mathrm{T} &= (\mathrm{M^{-1}L^3T^{-2}})^{\alpha}\mathrm{M}^{\beta}\mathrm{L}^{\gamma}\\ &= \mathrm{M}^{-\alpha+\beta}\mathrm{L}^{3\alpha+\gamma}\mathrm{T}^{-2\alpha}\end{aligned}$$

Equating powers of M, L and T with $\mathrm{M^0L^0T^1}$:

$$\begin{aligned}-\alpha + \beta &= 0\\ 3\alpha + \gamma &= 0\\ -2\alpha &= 1\end{aligned}$$

The equation $-2\alpha = 1$ gives

$$\alpha = -\frac{1}{2}$$

Then

$$\beta = \alpha = -\frac{1}{2}$$

and

$$\gamma = -3\alpha = \frac{3}{2}$$

Hence

$$\alpha = -\frac{1}{2}, \qquad \beta = -\frac{1}{2}, \qquad \gamma = \frac{3}{2}$$

3. A small probe is launched vertically upwards from the surface of a spherical moon. The radius of the moon is R metres, and the probe is at a distance r metres from the centre of the moon. The gravitational field strength g at distance r is modelled by

$$g = \frac{\Gamma}{r^2}$$

where Γ is a constant associated with the moon.

The probe has initial speed u at the surface and speed v when it reaches distance r. These speeds are related by

$$v^2 = u^2 - 2\Gamma\left(\frac{1}{R} - \frac{1}{r}\right)$$

The work done against gravity in moving a probe of mass m from the surface to distance r is

$$W = m\Gamma\left(\frac{1}{R} - \frac{1}{r}\right)$$

(a) Find the dimensions of Γ. **[2]**

(b) Use dimensional analysis to show that the equation for v^2 is dimensionally consistent. **[2]**

(c) Find the dimensions and SI units of W. **[2]**

Solution

(a) From

$$g = \frac{\Gamma}{r^2}$$

and since

$$[g] = \mathrm{LT}^{-2}, \qquad [r] = \mathrm{L}$$

we have

$$[\Gamma] = [g][r^2] = \mathrm{LT}^{-2} \times \mathrm{L}^2 = \mathrm{L}^3\mathrm{T}^{-2}$$

(b) In

$$v^2 = u^2 - 2\Gamma\left(\frac{1}{R} - \frac{1}{r}\right)$$

the dimensions of v^2 and u^2 are

$$[v^2] = [u^2] = (\mathrm{LT}^{-1})^2 = \mathrm{L}^2\mathrm{T}^{-2}$$

Also,

$$\left[\frac{1}{R}\right] = \left[\frac{1}{r}\right] = \mathrm{L}^{-1}$$

so

$$\left[\left(\frac{1}{R} - \frac{1}{r}\right)\right] = \mathrm{L}^{-1}$$

The factor 2 is dimensionless, so

$$\left[2\Gamma\left(\frac{1}{R}-\frac{1}{r}\right)\right] = [\Gamma]\mathrm{L}^{-1} = \mathrm{L}^3\mathrm{T}^{-2}\times\mathrm{L}^{-1} = \mathrm{L}^2\mathrm{T}^{-2}$$

Therefore every term in the equation has dimensions $\mathrm{L}^2\mathrm{T}^{-2}$, so the equation is dimensionally consistent.

(c) Using

$$W = m\Gamma\left(\frac{1}{R}-\frac{1}{r}\right)$$

with

$$[m] = \mathrm{M}, \qquad [\Gamma] = \mathrm{L}^3\mathrm{T}^{-2}, \qquad \left[\left(\frac{1}{R}-\frac{1}{r}\right)\right] = \mathrm{L}^{-1}$$

gives

$$[W] = \mathrm{M}\times\mathrm{L}^3\mathrm{T}^{-2}\times\mathrm{L}^{-1} = \mathrm{ML}^2\mathrm{T}^{-2}$$

Its SI units are

$$\mathrm{kg\,m^2\,s^{-2}}$$

which is the joule, J.

4. The increase in pressure p pascals between the surface and the bottom of a liquid of depth h metres and density $\rho\,\mathrm{kg\,m^{-3}}$ is modelled by

$$p = k\rho^{\alpha}h^{\beta}g$$

where $g\,\mathrm{m\,s^{-2}}$ is the acceleration due to gravity and k is a dimensionless constant.

(a) State the dimensions of p. **[1]**

(b) Use dimensional analysis to find the value of α and the value of β. **[3]**

Solution

(a) Pressure is force per unit area.

With [force] $= \mathrm{MLT^{-2}}$ and [area] $= \mathrm{L^2}$, this gives

$$[p] = \frac{\mathrm{MLT^{-2}}}{\mathrm{L^2}} = \mathrm{ML^{-1}T^{-2}}$$

(b) Using

$$p = k\rho^{\alpha}h^{\beta}g$$

and noting that k is dimensionless gives

$$[p] = [\rho]^{\alpha}[h]^{\beta}[g]$$

Now

$$[\rho] = \mathrm{ML^{-3}}, \qquad [h] = \mathrm{L}, \qquad [g] = \mathrm{LT^{-2}}$$

So the dimensions of the right-hand side are

$$\begin{aligned}(\mathrm{ML^{-3}})^{\alpha}(\mathrm{L})^{\beta}(\mathrm{LT^{-2}}) &= \mathrm{M^{\alpha}L^{-3\alpha}L^{\beta}LT^{-2}} \\ &= \mathrm{M^{\alpha}L^{-3\alpha+\beta+1}T^{-2}}\end{aligned}$$

But part (a) gives

$$[p] = \mathrm{ML^{-1}T^{-2}}$$

Equating powers of M, L and T:

$$\alpha = 1$$

and

$$-3\alpha + \beta + 1 = -1$$

Substituting $\alpha = 1$:

$$\begin{aligned}-3(1) + \beta + 1 &= -1 \\ \beta - 2 &= -1 \\ \beta &= 1\end{aligned}$$

Hence

$$\alpha = 1, \qquad \beta = 1$$

5. The force, F, exerted on a flat plate by a water jet is thought to depend on the density of the water, ρ, the cross-sectional area of the jet, A, and the speed of the water, v. You are given that ρ is measured in $\mathrm{kg\,m^{-3}}$.

(a) Find the dimensions of density ρ. **[1]**

A model of the force suggests the relationship

$$F = \frac{1}{2}\rho^{\alpha} A^{\beta} v^{\gamma}$$

(b) Explain whether or not it is possible to use dimensional analysis to verify that the constant $\frac{1}{2}$ is correct. **[1]**

(c) Use dimensional analysis to find the values of α, β and γ. **[4]**

Solution

(a) Density is mass per unit volume, so

$$[\rho] = \frac{\mathrm{M}}{\mathrm{L}^3} = \mathrm{ML}^{-3}$$

(b) No, it is not possible.

Dimensional analysis only checks that both sides of an equation have the same dimensions. The number $\frac{1}{2}$ is a dimensionless constant, so dimensional analysis cannot show whether $\frac{1}{2}$ is the correct numerical value.

(c) We use

$$F = \frac{1}{2}\rho^{\alpha} A^{\beta} v^{\gamma}$$

The factor $\frac{1}{2}$ is dimensionless, so it does not affect the dimensions.

Now write the dimensions of each quantity:

$$[F] = \mathrm{MLT}^{-2}, \qquad [\rho] = \mathrm{ML}^{-3}, \qquad [A] = \mathrm{L}^2, \qquad [v] = \mathrm{LT}^{-1}$$

So

$$\mathrm{MLT}^{-2} = (\mathrm{ML}^{-3})^{\alpha}(\mathrm{L}^2)^{\beta}(\mathrm{LT}^{-1})^{\gamma}$$

Combining powers gives

$$\mathrm{MLT}^{-2} = \mathrm{M}^{\alpha}\mathrm{L}^{-3\alpha+2\beta+\gamma}\mathrm{T}^{-\gamma}$$

Equating powers of M, L and T:

$$\begin{aligned} \text{For M}: &\quad \alpha = 1 \\ \text{For T}: &\quad -\gamma = -2 \implies \gamma = 2 \\ \text{For L}: &\quad -3\alpha + 2\beta + \gamma = 1 \end{aligned}$$

Substitute $\alpha = 1$ and $\gamma = 2$ into the L-equation:

$$\begin{aligned} -3(1) + 2\beta + 2 &= 1 \\ -1 + 2\beta &= 1 \\ 2\beta &= 2 \\ \beta &= 1 \end{aligned}$$

Hence

$$\alpha = 1, \qquad \beta = 1, \qquad \gamma = 2$$

So the model is

$$F = \frac{1}{2}\rho A v^2$$

6. A motor pulls a crate of mass m along a rough horizontal floor. The equation below is suggested to model the pulling force exerted by the motor, F, while the crate moves in a straight line.

$$F = k_1 mv\frac{\mathrm{d}v}{\mathrm{d}x} + k_2\mu mg + k_3\frac{Q}{x}$$

in which

- v is the speed of the crate when it has travelled distance x
- μ is the coefficient of friction between the crate and the floor
- g is the acceleration due to gravity
- Q is the total thermal energy transferred to the surroundings after the crate has moved distance x
- k_1, k_2 and k_3 are dimensionless constants

Determine whether the equation is dimensionally consistent. **[6]**

Solution

For the equation to be dimensionally consistent, every term must have the dimensions of force:

$$[F] = \mathrm{MLT}^{-2}$$

Also, k_1, k_2 and k_3 are dimensionless, and μ is dimensionless.

Now consider each term on the right-hand side:

$$[v] = \mathrm{LT}^{-1}, \qquad [x] = \mathrm{L}$$

so

$$\left[\frac{\mathrm{d}v}{\mathrm{d}x}\right] = \frac{[v]}{[x]} = \frac{\mathrm{LT}^{-1}}{\mathrm{L}} = \mathrm{T}^{-1}$$

For the first term:

$$\begin{aligned}\left[k_1 mv\frac{\mathrm{d}v}{\mathrm{d}x}\right] &= [m][v]\left[\frac{\mathrm{d}v}{\mathrm{d}x}\right] \\ &= \mathrm{M}\times\mathrm{LT}^{-1}\times\mathrm{T}^{-1} \\ &= \mathrm{MLT}^{-2}\end{aligned}$$

For the second term:

$$\begin{aligned}[k_2\mu mg] &= [m][g] \\ &= \mathrm{M}\times\mathrm{LT}^{-2} \\ &= \mathrm{MLT}^{-2}\end{aligned}$$

For the third term, Q is thermal energy transferred, so it has the dimensions of energy:

$$[Q] = \mathrm{ML}^2\mathrm{T}^{-2}$$

Therefore

$$\begin{aligned}\left[k_3\frac{Q}{x}\right] &= \frac{[Q]}{[x]} \\ &= \frac{\mathrm{ML^2T^{-2}}}{\mathrm{L}} \\ &= \mathrm{MLT^{-2}}\end{aligned}$$

Each term on the right-hand side has dimensions $\mathrm{MLT^{-2}}$, the same as force.

Therefore the equation is dimensionally consistent.

7. The speed of a particle falling vertically through a resistive medium is modelled by the formula

$$v = \frac{mg}{k}\left(1 - e^{-kt/m}\right)$$

where

v is the speed of the particle
m is the mass of the particle
g is the acceleration due to gravity
t is the time since the particle was released
k is a constant

Use dimensional analysis to determine the dimensions of k. **[2]**

Solution

For

$$v = \frac{mg}{k}\left(1 - e^{-kt/m}\right)$$

the factor $1 - e^{-kt/m}$ must be dimensionless.

In particular, the exponent of an exponential must have no dimensions, so

$$\left[\frac{kt}{m}\right] = 1$$

Using

$$[t] = \mathrm{T}, \qquad [m] = \mathrm{M}$$

we get

$$\frac{[k][t]}{[m]} = 1$$

so

$$\frac{[k]\mathrm{T}}{\mathrm{M}} = 1$$

Hence

$$[k] = \mathrm{MT}^{-1}$$

so the SI unit of k is $\mathrm{kg\,s^{-1}}$.

8. A comet of mass m measured in kg follows an elliptical orbit around a planet. The semi-major axis of the orbit is a measured in metres and the orbital period is T measured in seconds. The eccentricity e of the orbit is dimensionless.

A constant μ associated with the planet is defined by

$$T^2 = \frac{4\pi^2 a^3}{\mu}$$

and the angular momentum of the comet is given by

$$H = m\sqrt{\mu a(1-e^2)}$$

By using dimensional analysis, find the dimensions of:

(a) μ **[2]**

(b) H **[2]**

Solution

(a) In

$$T^2 = \frac{4\pi^2 a^3}{\mu}$$

the factor $4\pi^2$ is dimensionless, so

$$[T]^2 = \frac{[a]^3}{[\mu]}$$

Using $[T] = \mathrm{T}$ and $[a] = \mathrm{L}$ gives

$$\mathrm{T}^2 = \frac{\mathrm{L}^3}{[\mu]}$$

so

$$[\mu] = \frac{\mathrm{L}^3}{\mathrm{T}^2} = \mathrm{L}^3\mathrm{T}^{-2}$$

(b) The angular momentum is

$$H = m\sqrt{\mu a(1-e^2)}$$

Here $[m] = \mathrm{M}$, and since e is dimensionless, $(1-e^2)$ is also dimensionless.

Therefore

$$[H] = [m]\sqrt{[\mu][a]}$$

Using $[\mu] = \mathrm{L}^3\mathrm{T}^{-2}$ from part (a) and $[a] = \mathrm{L}$ gives

$$\begin{aligned}[H] &= \mathrm{M}\sqrt{(\mathrm{L}^3\mathrm{T}^{-2})(\mathrm{L})} \\ &= \mathrm{M}\sqrt{\mathrm{L}^4\mathrm{T}^{-2}} \\ &= \mathrm{ML}^2\mathrm{T}^{-1}\end{aligned}$$

9. A small ball of mass m kg is projected vertically upwards with speed $u\,\mathrm{m\,s^{-1}}$. During the motion, when its speed is $v\,\mathrm{m\,s^{-1}}$, it experiences an air resistance of magnitude kv^2 N, where k is a constant.

(a) Determine the dimensions of k. **[2]**

The greatest height reached by the ball is H m. A formula approximating H is

$$H = \frac{u^2}{2g} - \frac{ku^4}{4mg^2} + \frac{1}{6}k^2m^\alpha u^\beta g^\gamma$$

where $g\,\mathrm{m\,s^{-2}}$ is the acceleration due to gravity.

(b) Use dimensional analysis to find α, β and γ. **[4]**

Solution

(a) The air resistance has magnitude kv^2, and this is a force, so

$$[k][v]^2 = [F]$$

Now

$$[F] = \mathrm{MLT^{-2}}, \qquad [v] = \mathrm{LT^{-1}}$$

So

$$\begin{aligned}[k] &= \frac{[F]}{[v]^2} \\ &= \frac{\mathrm{MLT^{-2}}}{(\mathrm{LT^{-1}})^2} \\ &= \frac{\mathrm{MLT^{-2}}}{\mathrm{L^2T^{-2}}} \\ &= \mathrm{ML^{-1}}\end{aligned}$$

Hence the SI unit of k is $\mathrm{kg\,m^{-1}}$.

(b) Since H is a height,

$$[H] = \mathrm{L}$$

Therefore each term in

$$H = \frac{u^2}{2g} - \frac{ku^4}{4mg^2} + \frac{1}{6}k^2m^\alpha u^\beta g^\gamma$$

must have dimensions of length.

So the third term must satisfy

$$[k^2m^\alpha u^\beta g^\gamma] = \mathrm{L}$$

Using

$$[k] = \mathrm{ML^{-1}}, \qquad [m] = \mathrm{M}, \qquad [u] = \mathrm{LT^{-1}}, \qquad [g] = \mathrm{LT^{-2}}$$

we get

$$\begin{aligned}[k^2 m^\alpha u^\beta g^\gamma] &= (\mathrm{ML}^{-1})^2(\mathrm{M})^\alpha(\mathrm{LT}^{-1})^\beta(\mathrm{LT}^{-2})^\gamma \\ &= \mathrm{M}^{2+\alpha}\mathrm{L}^{-2+\beta+\gamma}\mathrm{T}^{-\beta-2\gamma}\end{aligned}$$

This must equal $\mathrm{M}^0\mathrm{L}^1\mathrm{T}^0$, so equating powers gives

$$\begin{aligned}2+\alpha &= 0 \\ -2+\beta+\gamma &= 1 \\ -\beta-2\gamma &= 0\end{aligned}$$

The first equation gives

$$\alpha = -2$$

The third equation gives

$$\beta = -2\gamma$$

Substitute into the second equation:

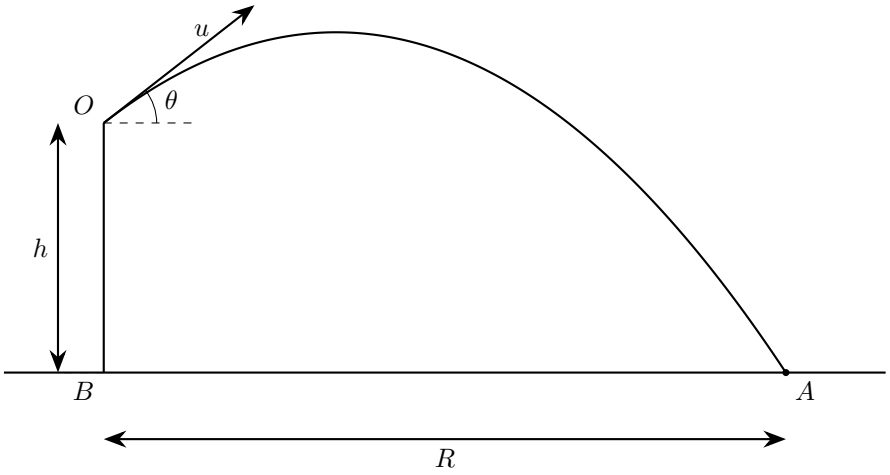
$$\begin{aligned}-2+(-2\gamma)+\gamma &= 1 \\ -2-\gamma &= 1 \\ -\gamma &= 3 \\ \gamma &= -3\end{aligned}$$

Then

$$\beta = -2(-3) = 6$$

Hence

$$\alpha = -2, \qquad \beta = 6, \qquad \gamma = -3$$



10. A particle is projected with speed u from a point O at the top of a cliff of height h above horizontal ground. The angle of projection is θ above the horizontal. The particle lands at the point A on the ground, and B is the foot of the cliff, so that $BA = R$, as shown in the diagram.

You are given that

$$R = \frac{u^\alpha \cos\theta}{g^\beta}\left(u^\gamma \sin\theta + \sqrt{u^\delta \sin^2\theta + 2gh}\right)$$

Use dimensional analysis to find α, β, γ and δ. **[4]**

Solution

The dimensions of the quantities are

$$[R] = \mathrm{L}, \qquad [u] = \mathrm{LT}^{-1}, \qquad [g] = \mathrm{LT}^{-2}, \qquad [h] = \mathrm{L}$$

and $\sin\theta$ and $\cos\theta$ are dimensionless.

First look at the expression inside the square root. Terms being added must have the same dimensions, so

$$[u^\delta \sin^2\theta] = [2gh]$$

Now

$$[2gh] = [g][h] = (\mathrm{LT}^{-2})(\mathrm{L}) = \mathrm{L}^2\mathrm{T}^{-2}$$

So

$$(\mathrm{LT}^{-1})^\delta = \mathrm{L}^2\mathrm{T}^{-2}$$

Equating powers of L and T gives

$$\delta = 2$$

Hence the whole square-root term has dimensions

$$\sqrt{\mathrm{L}^2\mathrm{T}^{-2}} = \mathrm{LT}^{-1}$$

Therefore $u^\gamma \sin\theta$ must also have dimensions LT^{-1}, so

$$(\mathrm{LT}^{-1})^\gamma = \mathrm{LT}^{-1}$$

which gives

$$\gamma = 1$$

Now the bracket

$$u^\gamma \sin\theta + \sqrt{u^\delta \sin^2\theta + 2gh}$$

has dimensions LT^{-1}. So

$$[R] = \left[\frac{u^\alpha \cos\theta}{g^\beta}\right]\left[u^\gamma \sin\theta + \sqrt{u^\delta \sin^2\theta + 2gh}\right]$$

$$\mathrm{L} = \frac{(\mathrm{LT}^{-1})^\alpha}{(\mathrm{LT}^{-2})^\beta}(\mathrm{LT}^{-1})$$

$$= \mathrm{L}^{\alpha-\beta+1}\mathrm{T}^{-\alpha+2\beta-1}$$

Equating powers of L and T with $\mathrm{L}^1\mathrm{T}^0$:

$$\alpha - \beta + 1 = 1, \qquad -\alpha + 2\beta - 1 = 0$$

The first equation gives

$$\alpha = \beta$$

Substitute into the second:

$$-\beta + 2\beta - 1 = 0$$

so

$$\beta = 1$$

and therefore

$$\alpha = 1$$

So the values are

$$\alpha = 1, \qquad \beta = 1, \qquad \gamma = 1, \qquad \delta = 2$$